FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

PART TEST – I

PAPER –1

TEST DATE: 20-11-2022

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. A, B, D

Sol. $-\dfrac{dx}{dt} = 0.1\sqrt{x}$

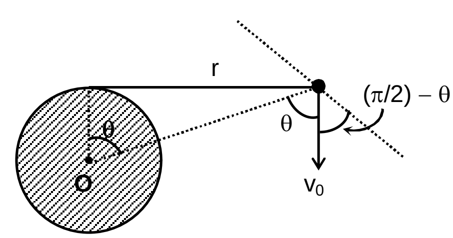
$$\int_9^0 \frac{dx}{\sqrt{x}} = -0.1\int_0^\tau dt \Rightarrow \tau = 60 \text{ sec}$$

$$\omega = \frac{0.1\sqrt{x}}{x} = \frac{0.1}{\sqrt{x}} \text{ rad/s}$$

$$a = v\omega = 0.1\sqrt{x}\frac{0.1}{\sqrt{x}} = 0.01 \text{ m/s}^2$$

2. A, C

Sol. Speed of the object will remain constant because tension is always perpendicular to the velocity.



So angular momentum about O = $mv_0 \sin\theta \dfrac{r}{\sin\theta} = mv_0 r$

$$\frac{mv_0 mv_0^2}{T_{max}} = 5 \text{ kg-m}^2\text{/sec}$$

3. B, C

Sol. For the bigger disc $C\omega_1 r_1^4 - \int f dt r_1 = 0$

For the smaller disc $C\omega_2 r_2^4 - \int f dt r_2 = 0$

So, $\omega_1 r_1^3 = \omega_2 r_2^3$ and $\dfrac{L_1}{L_2} = \dfrac{r_1}{r_2}$

4. A, C, D

Sol. In both situations $v_C = \frac{J}{m}$

In situation-(ii), $\omega = \frac{J}{m\ell}$

5. B, D

Sol. $H_B^{max} > H_A^{max} > H_C^{max}$ (as shown clearly)

So, $u_B^y > u_A^y > u_C^y$

Vertical component of initial velocity of each particle will decide the time taken.

So, $t_C < t_A < t_B$

6. B

Sol. For COM position, $m_1x_1 = m_2x_2$

Where m_1 and m_2 are masses of left and right part of the rod and x_1, x_2 are their respective position of centre of mass.

Simultaneously, $m_1x_1\omega = m_2x_2\omega$

and $I_1\omega \neq I_2\omega$

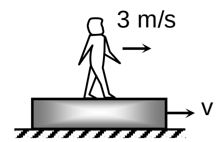
and $\frac{1}{2}I_1\omega^2 \neq \frac{1}{2}I_2\omega^2$

7. C

Sol. (I) 0 = 100v + 50(v + 3) (COLM)

v = –1 m/s

So, work done by man = $\frac{1}{2}(100)(1)^2 + \frac{1}{2}(50)(2)^2 = 150$ J



(II) $W_N + W_{mg} = \Delta$ K.E. (WET)

$W_N - \frac{1}{2}(2)(10)^2 = \frac{1}{2}(2)\left[(5)^2 - (15)^2\right]$

$W_N = -100$ J

(III) Unstable equilibrium points are x = 10 m, 30 m

Stable equilibrium point is x = 20 m

So, $U_i = 0$, and $U_f = -\frac{1}{100}(10)^4 = -100$ J

$W_{\text{conservative force}} = U_i - U_f = 100$ J

(IV) $W_{ext} = U_f - U_i = mg\left(\frac{5R}{8} - \frac{3R}{8}\right) = \frac{mgR}{4} = 100$ J

8. B

Sol. **In case-(I)**

$at = \omega x$

$y = \frac{1}{2}at^2$

$y = \frac{1}{2}a\frac{\omega^2x^2}{a^2}$

$y = \frac{\omega^2x^2}{2a}$

9. A

Sol. 3 kg block will be at rest. (so friction between S_2 and D = tension in the string connect to block D)
Considering A + B + C in a system
$T - 6 = 3a$, $20 - 2T = 2a_1$ and $a = 2a_1$
So, $a = \frac{8}{7}\ m/s^2$
So, there will be slipping between blocks A and B.
Now consider (B + C) in a system
$T' - 6 - 1 = 2a'$
$20 - 2T' = 2a_2$ and $a' = 2a_2$
So, $T' - 7 = 4a_2$
$20 - 2T' = 2a_2$
So, $a' = 1.2\ m/s^2$

10. C

Sol. **Case –I :** $\frac{mR^2}{2}\omega_0 = \left(\frac{mR^2}{2} + \frac{mR^2}{4} + \frac{mR^2}{4}\right)\omega$

$\omega = \frac{\omega_0}{2} = 3\ \text{rad/s}$

Case –II: $\omega = \omega_0 = 6\ \text{rad/s}$

Case –III: $\frac{mR^2}{2}\omega_0 + mv\frac{R}{2} = (mR^2)\omega$

So, $3 + \frac{8}{2} = \omega = 7\ \text{rad/s}$

Case –IV: $\frac{mR^2}{2}\omega_0 + mv\frac{R}{2}\left(\frac{1}{2}\right) = mR^2\omega$

$3 + \frac{8}{4} = \omega = 5\ \text{rad/s}$

Section – B

11. 4.50 (range 4.40 to 4.60)

Sol. (F) (2) = (N) (1) $\Rightarrow$ N = 2F (N is normal between rods)
and $(N)\left(\frac{1}{2}\right) = (F_1)(2)$
So, $F_1 = \frac{F}{2}$

12. 60.60 (range 60.00 to 61.00)

Sol. Total time taken = $\frac{0.01L}{0.0001v} = \frac{100L}{v}$

Total number of collision with moving wall in the said movement $= \frac{100L}{v\left(\frac{2L}{v}\right)} = 50$

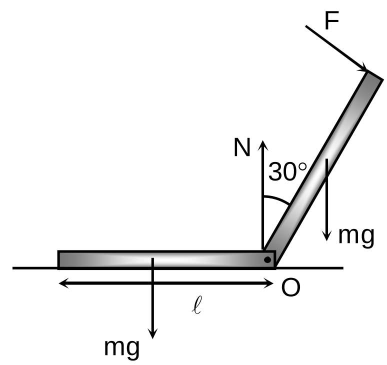
In each collision with the movable wall speed of the sphere increases by 0.0002 v.
So, final speed of sphere = v + 50 × 0.0002v = 1.01 v = 60.60 m/s

13. 2.50 (range 2.40 to 2.60)

Sol. from torque equilibrium about O

$$F\ell + mg\frac{\ell}{2}\frac{1}{2} = mg\frac{\ell}{2}$$

$$F = \frac{mg}{4}$$



14. 1.41 (range 1.35 to 1.45)

Sol. $\int N dt = mv\cos\theta \quad (v = \sqrt{2gh})$

So, $v_f = \dfrac{mv\sin\theta - \mu mv\cos\theta}{m} = \sqrt{2gh}\left(\dfrac{3}{5} - \dfrac{1}{2}\dfrac{4}{5}\right) = \dfrac{\sqrt{50}}{5} = \sqrt{2}$ m/s

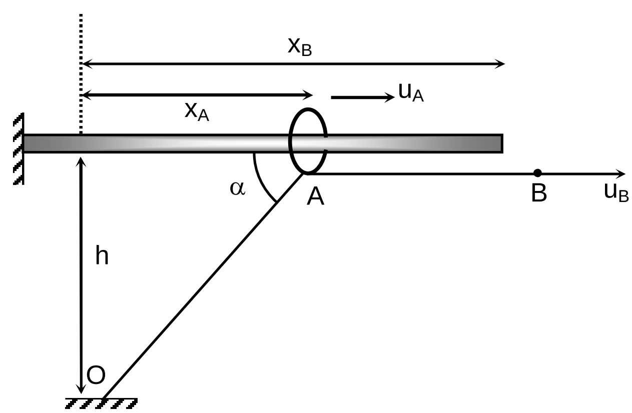
15. 7.46 (range 7.40 to 7.50)

Sol. $x_B - x_A + \sqrt{x_A^2 + h^2} = L$

$$\frac{dx_B}{dt} - \frac{dx_A}{dt} + \frac{x_A}{\sqrt{x_A^2 + h^2}}\frac{dx_A}{dt} = 0$$

So, $u_A = \dfrac{u_B}{1 - \cos\alpha} = \dfrac{1}{1 - \frac{\sqrt{3}}{2}} = \dfrac{2}{2 - \sqrt{3}}$

$\Rightarrow u_A = 7.46$ m/s



16. 2.43 (range 2.35 to 2.55)

Sol. $\Delta y_{CM} = \dfrac{1}{2}(10)\left(\dfrac{1}{10}\right)^2 = 0.05$ m

So, $h_{CM} = 1 + \dfrac{2}{3} - 0.05 = 1.62$ m

So, $y_{2m} = \dfrac{1.62 \times 3}{2} = 2.43$ m

17. 58.80 (range 57.00 to 59.00)

Sol. Assume the bloc M at rest, we see that both smaller cubes are applying net zero resultant horizontal force on the larger block.

So, N = Mg + mg ($\cos^2 37^\circ + \cos^2 53^\circ$) = Mg + mg

18. 2.00 (range 1.95 to 2.05)

Sol. $mg\cos 53^\circ = 0.5\text{ N}$

$mg\sin 53^\circ + N = m\omega^2 r$

So, $mg(0.8) + \dfrac{mg(0.6)}{0.5} = m\omega^2 r$

So, $g\left(\dfrac{8}{10} + \dfrac{6}{5}\right) = \omega^2 r$

So, $10 \times 2 = \omega^2 r$

So, $\omega = \sqrt{\dfrac{20}{5}} = 2.00\ \text{rad/s}$

Chemistry

PART – II

Section – A

19. B, D

Sol. $XeF_4 \rightarrow sp^3d^2 \rightarrow$ Square planar and nonplanar

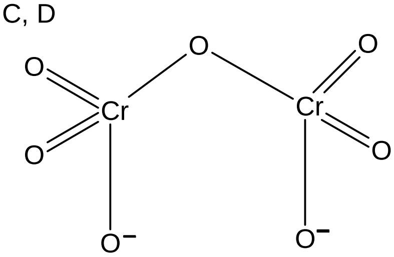
$H_2O_2 \rightarrow sp^3 \rightarrow$ open book structure. Nonplanar and polar

$XeF_5^{\ominus} \rightarrow$ Pentagonal planar and non – polar

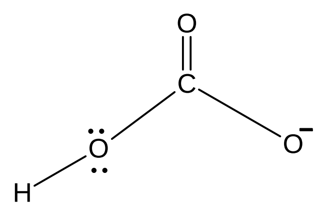
$XeF_5^{\oplus} \rightarrow$ Square pyramidal non – planar polar

20. C, D

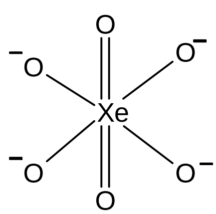
Sol.



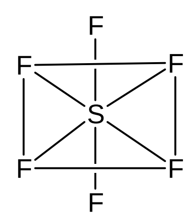
All Cr - O bonds are not equivalent



All C - O bonds are not equivalent



All Xe - O bonds are equivalent due to resonance



sp^3d^2 hybridisation

21. A, B, C

Sol. r = 0, R(r) = 0 so its true for p-orbital, Azimuthal quantum number is 1.

$$R(r) = 0, \frac{6r}{a^o} - \frac{r^2}{a_o^2} = 0$$

One radial node is observed at $r = 6a^o$ distance.
So, its true for 3p-orbital.
In P_x orbital nodal plane is in yz plane and p-orbital accommodate only 2-electron.

22. A, C, D

Sol. Tetrahedral will be formed by sp^3-hybridisation. Other three can be formed by distorted sp^3d hybridisation. i.e.

XeF_2 and I_3^- are sp^3d but linear shape.
BrF_3 is sp^3d but bent T-shaped.
SF_4 is sp^3d but see-saw shaped.

23. A, C

Sol. In CrO_8^{3-} Cr has +5 oxidation state.

KO_2 is also colorful.

24. A, B, C

Sol. In option (D) if we add inert gas at constant pressure and temperature. Equilibrium will get disturbed and more in backward direction.

25. A

Sol. For reaction $A \longrightarrow \text{Product}$

Using the given formula we can solve all the options

For zero order reaction $t = \dfrac{[A]_o - [A]_t}{k}$

For first order reaction $t = \dfrac{1}{k}\ell n\dfrac{[A]_o}{[A]_t}$

For second order reaction $t = \dfrac{1}{k}\left[\dfrac{1}{[A]_t} - \dfrac{1}{[A]_o}\right]$

26. B

Sol. Correct answer is I→P, R, S; II→P, S, T; III→P, R, S; IV→P, Q,R, S

27. C

Sol. Correct order is I→R,S; II→R; III→Q, S; IV→P

28. D

Sol. Correct answer is I→Q; II→P; III→S; IV→R

Section – B

29. 1.00

Sol. SiO_2 is neutral and Na has unit +ve charge AlO_2 should have unit – ve charge to balance the charge of molecule.

x(1) + y(– 1) + 136(0) + 250(0) = 0

x = y

$\dfrac{x}{y} = 1.00$

30. 9.09

Sol.

$$\begin{array}{llll} CH_3COOH & + \;\; NaOH & \longrightarrow CH_3COONa + H_2O \\ 0.60\ V & 0.20 \times 3V & \quad 0.60V \end{array}$$

$[CH_3COONa] = \dfrac{0.60V}{4V} = \dfrac{3}{20}M$

pH at 25°C for salt of WA + SB is

$pH = 7 + \dfrac{1}{2}(pK_a + \log(\text{salt}))$

$pH = 7 + \dfrac{1}{2}\left(5 + \log\dfrac{3}{20}\right)$

$= 7 + \dfrac{1}{2}(5 - 0.82)$

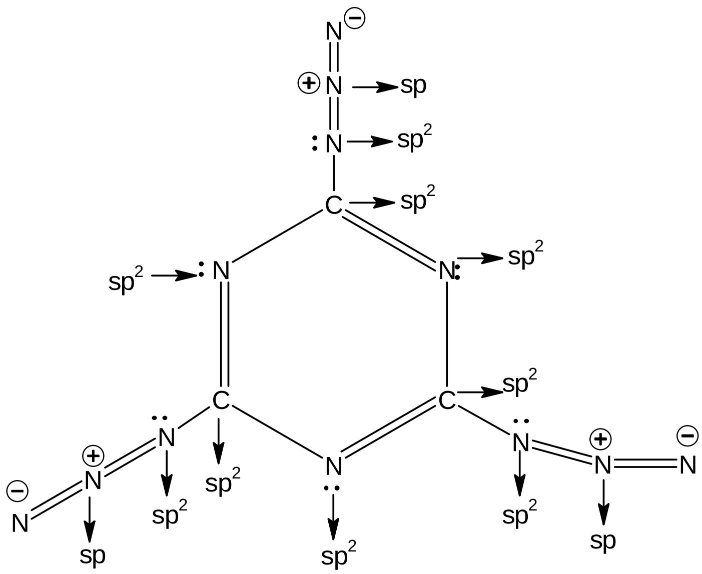
$= 7 + \dfrac{1}{2}(4.18)$

$= 7 + 2.09$

= 9.09

31. 1.50

Sol.



Maximum 6π bond can lies in same plane and sp^2 centre are 9.

$x = 9,\ y = 6$

$$\Rightarrow \frac{x}{y} = \frac{9}{6} = 1.50$$

32. 1.50

Sol. Balance reaction is

$$ClO_3^- + 6I^- + 6H_2SO_4 \longrightarrow 3I_2 + Cl^- + 6HSO_4^- + 3H_2O$$

x = 6, y = 6, z = 3

33. 6.93

Sol.

$$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$$

$2 - x \qquad x - y \qquad x$

$$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$$

$x - y \qquad \frac{y}{2} \qquad \frac{3y}{2}$

Given,

$2 - x = 1$ $\qquad \frac{3y}{2} = 0.75$

$x = 1$ $\qquad y = \left(\frac{1}{2}\right)$

$$x_1 = K_{C_1}\left(\frac{1}{2}\right).\left(\frac{1}{4}\right) = \left(\frac{1}{8}\right)$$

$$y_1 = K_{C_2} = \frac{\left(\frac{3}{8}\right)^{\frac{3}{2}}\left(\frac{1}{8}\right)^{\frac{1}{2}}}{\left(\frac{1}{4}\right)} = \left(\frac{3\sqrt{3}}{16}\right)$$

$a = [NH_3] = \frac{1}{4}M$

$[H_2S] = \frac{1}{2}M$

$b = [N_2] = \frac{1}{8}M$

$[H_2] = \frac{3}{8}M$

$\frac{y_1}{x_1(a+b)} = \frac{3\sqrt{3}}{16} \times \frac{8}{1} \times \frac{8}{3} = 4\sqrt{3}$

$= 4 \times 1.732$

= 6.928

= 6.93

34. 1.33

Sol. $K(\text{overall}) = \frac{K_1K_3}{K_2} = 10^{-4}\ \text{sec}^{-1}$

$A(\text{overall}) = \frac{A_1A_3}{A_2} = 2 \times 10^5\ \text{sec}^{-1}$

$E_a(\text{overall}) = E_{a_1} + E_{a_3} - E_{a_2} = 15\ \text{kJ}$

$\frac{K.A}{E_a} = \frac{2 \times 10^{-4} \times 10^5}{15} = 1.33$

35. 18.48

Sol. $MgCl_2 + 2NH_3 + 2H_2O \longrightarrow Mg(OH)_2(s)\downarrow + 2NH_4Cl$

$Mg(OH)_2(s) \rightleftharpoons Mg^{2+} + 2OH^-$

$(NH_4)_2SO_4$ is acidic so it will dissolved $Mg(OH)_2$ solid which is formed in mix. When all $Mg(OH)_2$ will re-dissolve the conc. of Mg^{2+} ions in the final solution will become equal to its initial conc.

$[NH_3]$ in mix initially $= \frac{0.70 \times 0.2}{1.0} = 0.14\ M$

$[Mg^{2+}]$ in mix initially $= \frac{0.30 \times 0.2}{1.0} = 0.06\ M$

When $[Mg^{2+}] = 0.06$

$[OH^-] = \sqrt{\frac{K_{SP}}{[Mg^{2+}]}} = \sqrt{\frac{6 \times 10^{-12}}{0.06}} = 10^{-5}\ M$

For NH_3 $K_b = \frac{[NH_4^+][OH^-]}{[NH_3]} \Rightarrow [NH_4^+] = \frac{K_b[NH_3]}{[OH^-]}$

$[NH_4^+] = \frac{2 \times 10^{-5} \times 0.14}{10^{-5}} = 0.28\ M$

Mole of $[NH_4^+] = 0.28$

Mole of $[NH_4]_2\,SO_4$ needed $= \frac{0.28}{2} = 0.14$

Mass of $[NH_4]_2\,SO_4 = 132 \times 0.14$

= 18.48 g.

36. 37.50

Sol.

$$A(aq) \longrightarrow 2B(aq) + C(aq)$$

	A	2B	C
$t = 0$	a	0	0
$t = 10$ min	$a - \frac{a}{2}$	a	$\frac{a}{2}$
$t = 20$ min	$a - \frac{3a}{4}$	$\frac{3a}{2}$	$\frac{3a}{4}$
$t = \infty$	0	$2a$	a

Let the normality of 'T' is N and initial volume of 'T' is used x ml

At t = 0 Meq. of A = Meq. of T

$a \times 2n = N \times x$ … (1)

At t = 10 min Meq. of (A + B + C) = Meq. of T

$\frac{a}{2} \times 2n + a \times n + \frac{a}{2} \times 2n = 40 \times N$ … (2)

At t = 20 min Meq. of (A + B + C) = Meq. of T

$\frac{a}{4} \times 2n + \frac{3a}{2} \times n + \frac{3a}{4} \times 2n = N \times y$ … (3)

At t = ∞ min Meq. of (B + C) = Meq. of T

$2a \times n + a \times 2n = N \times z$ … (4)

Solving these equilibrium $x = \frac{80}{3}\text{ml}, y = \frac{140}{3}\text{ml}, z = \frac{160}{3}\text{ml}$

Value of $\frac{y+z}{x} = 37.5$.

Mathematics

PART – III

Section – A

37. B, C, D

Sol. $f(x) = \dfrac{3}{\sqrt{2x^2+x} + \left(\dfrac{2\sin^2\theta}{\sqrt{2x^2+x}}\right)}$

Using AM $\geq$ GM

$\Rightarrow \dfrac{\sqrt{2x^2+x} + \dfrac{2\sin^2\theta}{\sqrt{2x^2+x}}}{2} \geq (2\sin^2\theta)^{\frac{1}{2}} \Rightarrow \dfrac{3}{2f(x)} \geq 2^{\frac{1}{2}}|\sin\theta| \Rightarrow f(x) \leq \dfrac{3}{2^{\frac{3}{2}}}|\text{cosec}\,\theta|$

$\because f(x) \leq \dfrac{3}{\sqrt{2}} \Rightarrow |\text{cosec}\,\theta| = 2 \Rightarrow |\sin\theta| = \dfrac{1}{2}$

38. A, B

Sol. Put x = 0 $\Rightarrow \int_0^1 f(t)e^{-t}dt = \dfrac{1}{a}$

$\Rightarrow \int_0^x f(t)dt = e^x - ae^{2x} \times \dfrac{1}{a} = e^x - 2^{2x}$, differentiating we get, $f(x) = e^x - 2e^{2x}$

39. A, B, C

Sol. (x – y) f(x + y) – (x + y) f(x – y)= 2y(x – y)(x + y)

Let x – y = u, x + y = v $\Rightarrow$ uf(v) – v(f(u) = (v – u)uv $\Rightarrow \dfrac{f(v)}{v} - v = \dfrac{f(u)}{u} - u = k$

$\Rightarrow$ f(x) = kx + x^2 $\because$ f(1) = 2 $\Rightarrow$ k = 1 $\Rightarrow$ f(x) = x^2 + x

40. D

Sol. $x^a y = \lambda^a$, on differentiating $ax^{a-1}y + x^a\dfrac{dy}{dx} = 0 \Rightarrow \dfrac{dy}{dx} = -\dfrac{ay}{x}$

Tangent at point (x_1, y_1) is $y - y_1 = -\dfrac{ay_1}{x_1}(x - x_1)$

Area = $\dfrac{1}{2}\lambda^a \cdot \dfrac{(a+1)^2}{a}(x_1)^{1-a}$ $\Rightarrow$ area is constant if a = 1

41. A, D

Sol. f(x) $\leq$ 0, F′(x) = f(x) as f(x) $\geq$ c F(x) $\Rightarrow$ F′(x) – c F(x) $\geq$ 0

$\Rightarrow e^{-cx}\cdot F'(x) - c\cdot e^{-cx}F(x) \geq 0 \Rightarrow \dfrac{d}{dx}\left(e^{-cx}\cdot F(x)\right) \geq 0$

$\Rightarrow e^{-cx}\cdot F(x)$ is an increasing function

$\Rightarrow e^{-cx}\cdot F(x) \geq e^{-c(0)}\cdot f(0)$ $\Rightarrow$ F(x) $\geq$ 0 $\Rightarrow$ f(x) $\geq$ 0 $\Rightarrow$ f(x) = 0

Also, $\dfrac{d}{dx}\left(e^{-x}g(x)\right) < 0 \Rightarrow e^x g(x)$ is a decreasing function

$\Rightarrow e^{-x}g(x) < e^{-(0)}g(0) \Rightarrow$ g(x) < 0 (as g(0) = 0)

Thus, f(x) = g(x) has one solution x = 0 also, $|x^2 + x - 6|$ = f(x) + g(x) = g(x) $\Rightarrow$ no solution

42. B, C

Sol. Solution of differential equation is $(\sin^2 x)y + (\sin x)y^2 + (\cos x + \cos^2 x) = 0$

43. D

Sol. (I) $\int_0^1 (f^4(x) - 8x^2 \cdot f^2(x) + 16x^4)dx \Rightarrow \int_0^1 (f^2(x) - 4x^2)^2 = 0 \Rightarrow f^2(x) = 4x^2$

(II) $f''(x) = -\frac{4x^3}{2\sqrt{1-x^4}} \Rightarrow f''(x) = 0$ at $x = 0$ and $f''(x)$ changes sign around it

(III) $g'(x) = x^2\left(\int_1^x f(u)du\right) + f(x)\cdot\left(\int_0^x t^2 dt\right)$ and $g'(1) = 1 \Rightarrow g''(1) = 6 + \frac{f'(1)}{3}$

(IV) $I_2 = -\frac{1}{3}\int_0^{-1} \frac{dt}{e^{-t}(2+t)} = -\frac{1}{3}\int_1^0 \frac{e^{t-1}}{1+t}dt = \frac{1}{3e}I_1$ (Let $-x^3 = t$)

44. C

Sol. (I) f(x) is discontinuous at $x = \frac{\pi}{2}, \pi$

(II) $A = \int_0^1 \left(x^{3/2} - \left(\frac{3x-1}{2}\right)\right)dx = \frac{3}{20}$

(III) Required area = $2\left|\int_1^2 x dx - \int_1^2 (x^2 - 2x + 2)dx\right| = \frac{1}{3}$

(IV) $I_n = \int_0^1 x^n\sqrt{1-x^2}dx = \int_0^1 x^{n-1}\cdot x\sqrt{1-x^2}dx = 0 + \frac{n-1}{3}\int_0^1 x^{n-2}(1-x^2)\sqrt{1-x^2}dx$

$= \frac{n-1}{3}I_{n-2} - \frac{n-1}{3}I_n \Rightarrow \frac{I_n}{I_{n-2}} = \frac{n-1}{n+2}$

45. B

Sol. (I) For $f(x) = x^3 - \frac{3}{2}x^2 + x + \frac{1}{4} \Rightarrow f(x) + f(1-x) = 1 \Rightarrow f(f(x)) + f(1-f(x)) = 1$

$\Rightarrow I = \int_{1/4}^{3/4} f(f(x))dx$ ..... (1)

$I = \int_{1/4}^{3/4} f(f(1-x))dx$ ..... (2)

Adding equation (1) and (2), we get $2I = \int_{1/4}^{3/4} f(f(x)) + f(1-f(x)) = \frac{1}{2}$ $\therefore I = \frac{1}{4}$

(II) Area $(A) = \int_0^1 (2x - x^2 - x^n)dx = \frac{1}{2} \Rightarrow n = 5$

(III) $f(x) + f(-x) = 1 \Rightarrow S = \frac{11}{2}$

(IV) $f(x) = \left(\frac{3}{2} - x^7\right)^{1/7}$ and $f(f(x)) = x$

46. D

Sol. (I) Put 2y = y $\Rightarrow$ f(x + y) = f(x) + f(y) + 2xy partially differentiate with respect to x
$f'(x + y) = f'(x) + 2y$ put x = 1, y = –1 $\Rightarrow f'(1) - f'(0) = 2$

(II) Let $\lim\limits_{x\to\infty} f(x) = \alpha$, then $\alpha + \frac{3\alpha - 1}{\alpha^2} = 3 \Rightarrow \alpha = 1$

(III) f(x) + f(x + π) = 2 $\therefore \frac{1}{2}f(x) + \frac{1}{2}f(x+\pi) = 1$ also af(x) + bf(x + c) = 1

On comparing a = b = $\frac{1}{2}$, c = π

(IV) $f(x + y) = 3^y f(x) + 2^x f(y)$ ..... (1)
Put x = 1 in equation (1) $f(1 + y) = 3^y f(1) + 2f(y)$
$y \to x \Rightarrow f(1 + x) = 3^x f(1) + 2f(x)$ ..... (2)
Now, put y = 1 in equation (1)
$f(1 + x) = 3f(x) + 2^x f(1)$ ..... (3)
From equation (2) and (3), we get $f(x) = 3^x - 2^x$

Section – B

47. 2023.00

Sol. $f(x) - f(y) \ge \ln x - \ln y + x - y$ ..... (1)
$x \to y$ and $y \to x$
$f(y) - f(x) \ge \ln y - \ln x + y - x$
$\Rightarrow f(x) - f(y) \le \ln x - \ln y + x - y$ ..... (2)
From equation (1) and (2), f(x) – f(y) = (ln x + x) – (ln y + y)
$\Rightarrow f(x) = \ln x + x \Rightarrow g(x) = 1 + \frac{1}{x} \Rightarrow g\left(\frac{1}{2022}\right) = 2023$

48. 5.00

Sol. $\sin 3\theta = 3\sin\theta - 4\sin^3\theta = \sin\theta(1 + 2\cos 2\theta) \Rightarrow \sin x = \sin\frac{x}{3}\left(1 + 2\cos\frac{2x}{3}\right)$

$\sin\frac{x}{3} = \sin\left(\frac{x}{3^2}\right)\left(1 + 2\cos\frac{2x}{3^2}\right)$ and so on multiplying all

$$\sin x = \sin\left(\frac{x}{3^n}\right)\prod_{r=1}^{n}\left(1 + 2\cos\frac{2x}{3^r}\right) \Rightarrow \frac{\sin x}{x} = \lim_{n\to\infty}\frac{\sin\left(\frac{x}{3^n}\right)}{\left(\frac{x}{3^n}\right)} \cdot \prod_{r=1}^{n}\frac{\left(1 + 2\cos\frac{2x}{3^r}\right)}{3} \Rightarrow f(x) = \frac{\sin x}{x}$$

Thus |x f(x)| + ||x – 2| – 1| = |sin x| + ||x – 2| – 1| is not differentiable at 5 points in (0, 3π)

49. 4.50

Sol. $$I = \int_0^2 (3x^2 - 6x + 4)\cos(x^3 - 3x^2 + 4x - 2)dx + \int_0^2 (3x - 3)\cos(x^3 - 3x^2 + 4x - 2)dx$$
Let $I = I_1 + I_2$
$I_1 = \int_{-2}^{2}(\cos t)dt = 2\sin 2$ (where $t = x^3 - 3x^2 + 4x - 2$)
$I_2 = 3\int_{-1}^{1} u\cos(u^3 + u)du = 0$ (where u = x – 1)

50. 7.63

Sol. If each of two positive real number x, y is not integer but x + y is an integer,
then [x] + [y] = x + y – 1

$\left(\because \{x\}+\{y\}=1\right)$

Since, $\dfrac{305k}{503}+\dfrac{305(503-k)}{503}=305$ for $1\le k\le 502$

$\Rightarrow$ $S=\left[\dfrac{305}{503}\right]+\left[\dfrac{305\cdot 2}{503}\right]+.....+\left[\dfrac{305\cdot 502}{503}\right]$ = 304.251 = 76304

51. 4044.00

Sol. $I_2=\int\limits_0^1 x^{1010}(1-x^{2022})^{1010}dx$ let $x^{1011}=t$

$\Rightarrow I_2=\dfrac{1}{1011}\int\limits_0^1 (1-t^2)^{1010}dt = \dfrac{1}{1011}\int\limits_0^1\left(1-(1-t^2)\right)^{1010}dt = \dfrac{1}{1011}\int\limits_0^1 t^{1010}(2-t)^{1010}dt$

Put t = 2y

$I_2=\dfrac{2^{2021}}{1011}\int\limits_0^{1/2} y^{1010}(1-y)^{1010}dy$ and $I_1=\int\limits_0^1 x^{1010}(1-x)^{1010}dx = 2\int\limits_0^{1/2} x^{1010}(1-x)^{1010}dx$

$\Rightarrow I=2^{2022}\dfrac{I_1}{I_2}=4044$

52. 199.00

Sol. $f(x)=\ln x-\dfrac{1}{x}\Rightarrow f(e^{200})=200-\dfrac{1}{e^{200}}$

53. 16.00

Sol. $A_1=4\int\limits_0^{\pi/4}(\cos x-\sin x)dx=4\left(\sqrt{2}-1\right)$ also $A_1=A_2=\dfrac{A}{2}$

54. 4.75

Sol. $I=\dfrac{1}{2}\int\limits_0^{\pi/2}\sin x\left(\ln(\sin^2 x)\right)dx$, put cos x = t

$I=\dfrac{1}{2}\int\limits_0^1 \ln(1-t^2)dt = \dfrac{1}{2}\int\limits_0^1\left(-t^2-\dfrac{(-t^2)^2}{2}+\dfrac{(-t^2)^3}{3}+.....\right)dt = -\dfrac{1}{2}\left[\dfrac{1}{3}+\dfrac{1}{10}+\dfrac{1}{21}+.....\right]$

$= -\left[\dfrac{1}{6}+\dfrac{1}{20}+\dfrac{1}{42}+.....\right]=-\left[\left(\dfrac{1}{2}-\dfrac{1}{3}\right)+\left(\dfrac{1}{4}-\dfrac{1}{5}\right)+.....\right] = \log_e 2-1$

and $A=\dfrac{15}{4}+\ln 2$